1. What is the .NET Framework primarily used for?
   - A) Designing graphics
   - B) Building and running applications
   - C) Managing databases
   - D) Writing operating systems
   - **Answer: B) Building and running applications**
2. Which feature in Visual Studio 2022 helps developers identify and fix code issues in real-time?
   - A) IntelliSense
   - B) Code Lens
   - C) Live Share
   - D) Debugging
   - **Answer: A) IntelliSense**
3. What is one of the key performance improvements in Visual Studio 2022?
   - A) Faster IDE load times
   - B) Improved code syntax coloring
   - C) Enhanced audio support
   - D) Built-in game development tools
   - **Answer: A) Faster IDE load times**
4. What are templates in Visual Studio 2022 used for?
   - A) Creating new project types
   - B) Changing color themes
   - C) Managing database connections
   - D) Optimizing application performance
   - **Answer: A) Creating new project types**
5. Which of the following is not a fundamental data type in C#?
   - A) int
   - B) bool
   - C) string
   - D) image
   - **Answer: D) image**
6. What does the ‘+’ operator do in C#?
   - A) Subtracts two numbers
   - B) Multiplies two numbers
   - C) Adds two numbers
   - D) Divides two numbers
   - **Answer: C) Adds two numbers**
7. Which operator is used for logical AND in C#?
   - A) ||
   - B) &&
   - C) !

- D) ==
- **Answer: B) &&**

8. How do you declare an integer variable named "age" in C#?
    - A) int age;
    - B) integer age;
    - C) var age;
    - D) age int;
    - **Answer: A) int age;**

9. How do you access a method named "Display" from an object "obj" in C#?
    - A) obj->Display();
    - B) obj.Display();
    - C) obj:Display();
    - D) obj.Display;
    - **Answer: B) obj.Display();**

10. Which of the following is a correct way to cast an integer "num" to a double in C#?
    - A) double num = (double)num;
    - B) double num = num;
    - C) double num = (num)double;
    - D) double num = (cast)num;
    - **Answer: A) double num = (double)num;**

11. How do you concatenate two strings "str1" and "str2" in C#?
    - A) str1.concat(str2);
    - B) str1 & str2;
    - C) str1 + str2;
    - D) str1 && str2;
    - **Answer: C) str1 + str2;**

12. Which tool do you use to create a new .NET Framework application in Visual Studio 2022?
    - A) SQL Server
    - B) Project Explorer
    - C) Solution Explorer
    - D) Template Browser
    - **Answer: D) Template Browser**

13. What keyword is used in C# to create a conditional statement?
    - A) case
    - B) if
    - C) switch
    - D) select
    - **Answer: B) if**

14. Which of the following loops is not available in C#?

- A) while
- B) for
- C) do-while
- D) until
- **Answer: D) until**

15. How do you declare an array of integers named "numbers" with five elements in C#?

- A) int[] numbers = new int[5];
- B) int numbers = int[5];
- C) var numbers = int[5];
- D) array numbers = new int[5];
- **Answer: A) int[] numbers = new int[5];**

16. Which statement is used to include a namespace in a C# program?

- A) include
- B) import
- C) use
- D) using
- **Answer: D) using**

17. What is the purpose of a breakpoint in Visual Studio?

- A) To optimize code performance
- B) To temporarily stop program execution for debugging
- C) To format code
- D) To generate documentation
- **Answer: B) To temporarily stop program execution for debugging**

18. How do you set a breakpoint in Visual Studio 2022?

- A) Double-click on the code line
- B) Right-click and select "Set Breakpoint"
- C) Click on the left margin next to the code line
- D) Select "Debug" from the menu and choose "Add Breakpoint"
- **Answer: C) Click on the left margin next to the code line**

19. What happens when code execution reaches a breakpoint?

- A) The program continues running
- B) The program crashes
- C) The program stops, allowing for inspection
- D) The program restarts
- **Answer: C) The program stops, allowing for inspection**

20. Which window in Visual Studio allows you to view variables and their values during a breakpoint?

- A) Solution Explorer
- B) Output Window
- C) Watch Window

- D) Error List
- **Answer: C) Watch Window**

1. Which of the following correctly declares and initializes an array of integers in C#?
   - A) int[] numbers = {1, 2, 3};
   - B) int numbers = {1, 2, 3};
   - C) int[] numbers = new {1, 2, 3};
   - D) int numbers[] = new int[3];
   - **Answer: A) int[] numbers = {1, 2, 3};**
2. What type of loop would you use to iterate through an array in C#?
   - A) if-else loop
   - B) while loop
   - C) for loop
   - D) switch loop
   - **Answer: C) for loop**
3. What best describes a method in programming?
   - A) A variable that stores data
   - B) A function that performs a specific task
   - C) A loop that iterates through values
   - D) A conditional statement
   - **Answer: B) A function that performs a specific task**
4. How do you define a method in C# that takes no parameters and returns no value?
   - A) void MyMethod() {}
   - B) def MyMethod() {}
   - C) function MyMethod() {}
   - D) var MyMethod = function() {}
   - **Answer: A) void MyMethod() {}**
5. How do you call a method named "Calculate" from an object "calc"?
   - A) calc.Calculate();
   - B) Calculate.calc();
   - C) calc->Calculate();
   - D) call calc.Calculate();
   - **Answer: A) calc.Calculate();**
6. Which Visual Studio feature allows you to step through your code line by line to debug it?
   - A) Code Lens
   - B) IntelliSense
   - C) Step Over
   - D) Breakpoints
   - **Answer: D) Breakpoints**

7. How do you define an optional parameter in a C# method?

- A) void MyMethod(int a = 10) {}
- B) void MyMethod(int a?) {}
- C) void MyMethod(int optional a) {}
- D) void MyMethod(optional int a) {}
- **Answer: A) void MyMethod(int a = 10) {}

8. What keyword is used to specify an output parameter in a method?

- A) out
- B) ref
- C) opt
- D) yield
- **Answer: A) out**

9. What does the & operator do in C#?

- A) Logical AND
- B) Bitwise AND
- C) Logical OR
- D) Bitwise OR
- **Answer: B) Bitwise AND**

10. Which of the following evaluates to true in Boolean arithmetic?

- A) true && false
- B) true || false
- C) false && false
- D) false || false
- **Answer: B) true || false**

11. What best describes an exception in programming?

- A) An unusual but acceptable behavior
- B) A runtime error that can be handled
- C) A syntax error
- D) A logic error that crashes the program
- **Answer: B) A runtime error that can be handled**

12. Which keyword is used to handle exceptions in C#?

- A) catch
- B) handle
- C) error
- D) rescue
- **Answer: A) catch**

13. What is the purpose of a finally block in exception handling?

- A) To define code that executes only if no exception occurs
- B) To define code that always executes, regardless of an exception
- C) To catch exceptions that are not handled by previous catch blocks
- D) To log exceptions

- **Answer: B) To define code that always executes, regardless of an exc eption**

14. How do you throw an exception in C#?
- A) throw new Exception("Error message");
- B) raise new Exception("Error message");
- C) throw Exception("Error message");
- D) raise Exception("Error message");
- **Answer: A) throw new Exception("Error message");**

15. What is logging primarily used for in software development?
- A) To generate reports
- B) To keep a record of application events and errors
- C) To format code
- D) To debug syntax errors
- **Answer: B) To keep a record of application events and errors**

16. What is the purpose of application profiling?
- A) To analyze the performance of an application
- B) To document the application
- C) To update the user interface
- D) To create backups
- **Answer: A) To analyze the performance of an application**

17. Which tool is used to monitor the performance of a .NET application?
- A) Solution Explorer
- B) Task Manager
- C) Performance Counters
- D) IntelliSense
- **Answer: C) Performance Counters**

18. Why is it important to organize your code before implementing Object-Oriented Programming (OOP)?
- A) To ensure better readability and maintainability
- B) To reduce the size of the codebase
- C) To speed up code compilation
- D) To improve syntax highlighting
- **Answer: A) To ensure better readability and maintainability**

19. What is the primary purpose of using classes in OOP?
- A) To define methods
- B) To encapsulate data and behavior
- C) To execute conditional logic
- D) To create loops
- **Answer: B) To encapsulate data and behavior**

20. How do you instantiate an object of a class named "Car" in C#?
- A) Car myCar = new Car();

- B) Car myCar = Car();
- C) Car myCar = Car.new();
- D) Car myCar = create Car();
- **Answer: A) Car myCar = new Car();**

1. What is inheritance in object-oriented programming?
   - A) A process of defining a new class based on an existing class
   - B) A method of data encapsulation
   - C) A technique to manage memory
   - D) A way to hide data
   - **Answer: A) A process of defining a new class based on an existing class**
2. How do we denote inheritance in C#?
   - A) Using the colon (:) symbol
   - B) Using the plus (+) symbol
   - C) Using the arrow (->) symbol
   - D) Using the equals (=) symbol
   - **Answer: A) Using the colon (:) symbol**
3. What is one advantage of using inheritance?
   - A) Reduced code readability
   - B) Code reusability
   - C) Increased complexity
   - D) Difficulty in maintenance
   - **Answer: B) Code reusability**
4. What is a disadvantage of inheritance?
   - A) Simplified code structure
   - B) Single level of hierarchy
   - C) Increased coupling between classes
   - D) Easier to maintain code
   - **Answer: C) Increased coupling between classes**
5. What is an abstract class in C#?
   - A) A class that cannot be instantiated and is meant to be inherited
   - B) A class that cannot have methods
   - C) A class that must be instantiated
   - D) A class that does not have any members
   - **Answer: A) A class that cannot be instantiated and is meant to be inherited**
6. What is the purpose of a sealed class in C#?
   - A) To allow inheritance from the class
   - B) To prevent the class from being inherited
   - C) To make the class abstract
   - D) To hide the class members

- **Answer: B) To prevent the class from being inherited**

7. What is polymorphism in object-oriented programming?
   - A) The ability of different objects to respond to the same method call in different ways
   - B) The ability to create objects
   - C) The ability to encapsulate data
   - D) The ability to inherit properties
   - **Answer: A) The ability of different objects to respond to the same method call in different ways**

8. What does it mean to overload a method in C#?
   - A) To define multiple methods with the same name but different parameters
   - B) To define multiple methods with different names but the same parameters
   - C) To define a single method that performs multiple tasks
   - D) To define a method that takes no parameters
   - **Answer: A) To define multiple methods with the same name but different parameters**

9. Why do we use method overloading?
   - A) To increase the complexity of the code
   - B) To provide multiple ways to call a method with different parameters
   - C) To reduce the number of methods in a class
   - D) To enforce a single method signature
   - **Answer: B) To provide multiple ways to call a method with different parameters**

10. Which of the following is an example of an overloaded constructor in C#?
    - A) public MyClass() {} and public MyClass(int a) {}
    - B) public void MyClass() {} and public void MyClass(int a) {}
    - C) public MyClass() {} and public MyClass() {}
    - D) public MyClass(int a, int b) {} and public MyClass(int c) {}
    - **Answer: A) public MyClass() {} and public MyClass(int a) **

11. What does it mean to override a method in C#?
    - A) To provide a new implementation for a method in a derived class that exists in the base class
    - B) To define multiple methods with the same name
    - C) To hide a method in the base class
    - D) To create a method with the same name as a variable
    - **Answer: A) To provide a new implementation for a method in a derived class that exists in the base class**

12. Why do we use method overriding?
    - A) To change the behavior of a method in a derived class

- B) To remove a method from a class
- C) To add new methods to a class
- D) To merge two methods into one
- **Answer: A) To change the behavior of a method in a derived class**

13. What is an abstract method in C#?

- A) A method that has no implementation and must be overridden in derived classes
- B) A method that is private and cannot be accessed
- C) A method that can be instantiated
- D) A method that is sealed
- **Answer: A) A method that has no implementation and must be overridden in derived classes**

14. What is an interface in C#?

- A) A blueprint for a class that defines methods without implementation
- B) A class that contains only private members
- C) A class that cannot be inherited
- D) A class that has all methods implemented
- **Answer: A) A blueprint for a class that defines methods without implementation**

15. How do you implement an interface in a class in C#?

- A) Using the keyword "implement"
- B) Using the keyword "interface"
- C) Using the colon (:) symbol
- D) Using the equals (=) symbol
- **Answer: C) Using the colon (:) symbol**

16. Why do we use interfaces?

- A) To provide a way to achieve multiple inheritance
- B) To encapsulate data
- C) To reduce code readability
- D) To prevent code reuse
- **Answer: A) To provide a way to achieve multiple inheritance**

17. Which keyword is used to define an abstract method in an abstract class?

- A) virtual
- B) override
- C) abstract
- D) sealed
- **Answer: C) abstract**

18. Which keyword is used to prevent a class from being inherited?

- A) abstract
- B) sealed
- C) virtual

- D) override
- **Answer: B) sealed**

19. Which keyword is used to provide a new implementation for a method in a derived class?

- A) virtual
- B) abstract
- C) new
- D) override
- **Answer: D) override**

20. Which keyword is used to define an interface in C#?

- A) class
- B) struct
- C) interface
- D) enum
- **Answer: C) interface**

1. What is a key benefit of inheriting from .NET Framework classes?

- A) Reduces memory usage
- B) Allows reuse of existing functionality
- C) Increases program execution speed
- D) Simplifies code readability
- **Answer: B) Allows reuse of existing functionality**

2. Which base class do most .NET classes inherit from?

- A) **System.Object**
- B) **System.Base**
- C) **System.Class**
- D) **System.Type**
- **Answer: A) System.Object**

3. How do you define a custom exception in C#?

- A) class MyCustomException : Exception {}
- B) class MyCustomException : Error {}
- C) class MyCustomException : Throwable {}
- D) class MyCustomException : System {}
- **Answer: A) class MyCustomException : Exception {}**

4. Which keyword is used to throw an exception in C#?

- A) raise
- B) throw
- C) error
- D) catch
- **Answer: B) throw**

5. How do you catch a custom exception in a try-catch block?

- A) try { ... } catch (MyCustomException ex) { ... }

- B) try { ... } catch (Exception ex) { ... }
- C) try { ... } finally (MyCustomException ex) { ... }
- D) try { ... } catch (Error ex) { ... }
- **Answer: A) try { ... } catch (MyCustomException ex) { ... }**

6. What is a generic type in C#?
   - A) A type that allows code reuse with different data types
   - B) A type that is specific to integers
   - C) A type that only works with strings
   - D) A type that cannot be inherited
   - **Answer: A) A type that allows code reuse with different data types**
7. What is an extension method in C#?
   - A) A method that extends the functionality of a class
   - B) A method that changes the base class
   - C) A method that overrides another method
   - D) A method that inherits from a base class
   - **Answer: A) A method that extends the functionality of a class**
8. Why are generics used in C#?
   - A) To increase code duplication
   - B) To enable type safety and code reuse
   - C) To make code harder to read
   - D) To restrict the use of different data types
   - **Answer: B) To enable type safety and code reuse**
9. What is an advantage of using generics in C#?
   - A) Reduced code readability
   - B) Increased type safety
   - C) Slower execution time
   - D) Increased memory usage
   - **Answer: B) Increased type safety**
10. What is the purpose of constraining generics in C#?
    - A) To limit the types that can be used with a generic class or method
    - B) To increase the flexibility of generics
    - C) To make code less efficient
    - D) To prevent inheritance
    - **Answer: A) To limit the types that can be used with a generic class or method**
11. Which namespace must be included to use generic collections in C#?
    - A) **System.Text**
    - B) **System.IO**
    - C) **System.Collections**
    - D) **System.Collections.Generic**
    - **Answer: D) System.Collections.Generic**

12. How do you declare a generic List of integers in C#?

- A) List<int> myList = new List<int>();
- B) ArrayList myList = new ArrayList();
- C) List<int> myList = new List();
- D) List myList = new List<int>();
- **Answer: A) List<int> myList = new List<int>();**

13. What is a key feature of a generic Dictionary in C#?

- A) It stores elements as key-value pairs
- B) It allows duplicate keys
- C) It does not maintain any order
- D) It is a single-dimensional array
- **Answer: A) It stores elements as key-value pairs**

14. How do you declare a generic Dictionary with string keys and integer values in C#?

- A) Dictionary<string, int> myDict = new Dictionary<string, int>();
- B) Dictionary<string, int> myDict = new Dictionary();
- C) Hashtable myDict = new Hashtable();
- D) ArrayList myDict = new ArrayList();
- **Answer: A) Dictionary<string, int> myDict = new Dictionary<string, int>();**

15. Which interface is commonly implemented by collections in C#?

- A) IComparable
- B) IDisposable
- C) ICollection
- D) ISerializable
- **Answer: C) ICollection**

16. What does implementing the IEnumerable interface allow a collection to do?

- A) Store elements in a dictionary
- B) Be used in a foreach loop
- C) Maintain a fixed size
- D) Implement custom sorting algorithms
- **Answer: B) Be used in a foreach loop**

17. Which method must be implemented when a class implements the IEnumerable interface?

- A) Add()
- B) GetEnumerator()
- C) Remove()
- D) Contains()
- **Answer: B) GetEnumerator()**

18. Why might you create a custom exception in C#?

- A) To create new data types
- B) To handle specific error conditions unique to your application
- C) To improve code performance
- D) To avoid using the try-catch block
- **Answer: B) To handle specific error conditions unique to your application**

19. What is a key benefit of catching custom exceptions?

- A) Reduced code flexibility
- B) Improved code readability and maintainability
- C) Increased application size
- D) Decreased code performance
- **Answer: B) Improved code readability and maintainability**

20. Which keyword is used to define a custom exception class in C#?

- A) exception
- B) custom
- C) class
- D) Exception
- **Answer: C) class**

1. Which class in .NET is used to read from and write to files?

- A) StreamReader
- B) File
- C) FileStream
- D) Directory
- **Answer: B) File**

2. What method is used to check if a file exists in C#?

- A) **File.Exists**
- B) **File.Check**
- C) **File.Find**
- D) **File.Search**
- **Answer: A) File.Exists**

3. Which method is used to delete a file in C#?

- A) **File.Remove**
- B) **File.Delete**
- C) **File.Destroy**
- D) **File.Clear**
- **Answer: B) File.Delete**

4. Which class is used to create, delete, and manipulate directories?

- A) Directory
- B) File
- C) FileSystem
- D) DirectoryInfo

- **Answer: A) Directory**

5. How do you create a new directory in C#?
   - A) **Directory.Create**
   - B) **Directory.New**
   - C) **Directory.CreateDirectory**
   - D) **Directory.NewDirectory**
   - **Answer: C) Directory.CreateDirectory**
6. What method is used to combine two parts of a file path in C#?
   - A) **Path.Merge**
   - B) **Path.Combine**
   - C) **Path.Unite**
   - D) **Path.Join**
   - **Answer: B) Path.Combine**
7. What is the purpose of serialization in .NET?
   - A) To increase code readability
   - B) To convert an object into a format that can be easily stored or transmitted
   - C) To compress files
   - D) To encrypt data
   - **Answer: B) To convert an object into a format that can be easily stored or transmitted**
8. Which attribute is used to mark a class as serializable in C#?
   - A) [Serializable]
   - B) [SerializableClass]
   - C) [Serialized]
   - D) [Serialization]
   - **Answer: A) [Serializable]**
9. Which formatter is used to serialize objects into binary format in C#?
   - A) BinarySerializer
   - B) BinaryFormatter
   - C) BinaryWriter
   - D) BinaryConverter
   - **Answer: B) BinaryFormatter**
10. What is used to serialize objects into XML format in C#?
    - A) XmlSerializer
    - B) XmlWriter
    - C) XmlConverter
    - D) XmlFormatter
    - **Answer: A) XmlSerializer**
11. Which library is commonly used for JSON serialization in C#?
    - A) **Newtonsoft.Json**

- B) JsonSerializer
- C) JsonWriter
- D) JsonConverter
- **Answer: A) Newtonsoft.Json**

12. Why would you create a custom serializer in C#?

- A) To enhance code readability
- B) To handle custom serialization logic that default serializers can't handle
- C) To reduce data size
- D) To avoid using third-party libraries
- **Answer: B) To handle custom serialization logic that default serializers can't handle**

13. What is a stream in the context of .NET?

- A) A collection of files
- B) An object used to read and write data sequentially
- C) A type of array
- D) A method to manipulate directories
- **Answer: B) An object used to read and write data sequentially**

14. Which of the following is not a type of stream in the .NET Framework?

- A) FileStream
- B) MemoryStream
- C) DataStream
- D) NetworkStream
- **Answer: C) DataStream**

15. Which class is used to read binary data from a stream?

- A) StreamReader
- B) BinaryReader
- C) FileReader
- D) DataReader
- **Answer: B) BinaryReader**

16. How do you write binary data to a stream in C#?

- A) BinaryWriter.Write(data)
- B) StreamWriter.Write(data)
- C) FileWriter.Write(data)
- D) DataWriter.Write(data)
- **Answer: A) BinaryWriter.Write(data)**

17. Which class is used to read text data from a stream?

- A) StreamReader
- B) TextReader
- C) FileReader
- D) DataReader
- **Answer: A) StreamReader**

18. How do you write text data to a stream in C#?

- A) TextWriter.Write(data)
- B) StreamWriter.Write(data)
- C) FileWriter.Write(data)
- D) DataWriter.Write(data)
- **Answer: B) StreamWriter.Write(data)**

19. Which class provides methods for manipulating file and directory paths in C#?

- A) Path
- B) Directory
- C) File
- D) Stream
- **Answer: A) Path**

20. What is the opposite process of serialization?

- A) Encryption
- B) Deserialization
- C) Decryption
- D) Compression
- **Answer: B) Deserialization**

1. What is the primary purpose of the **ADO.NET** Entity Framework?

- A) To manage database security
- B) To enable developers to work with data using objects
- C) To design the user interface
- D) To perform file operations
- **Answer: B) To enable developers to work with data using objects**

2. Which tool is used to create an Entity Data Model in Visual Studio?

- A) SQL Server Management Studio
- B) Entity Data Model Wizard
- C) LINQPad
- D) Visual Studio Code
- **Answer: B) Entity Data Model Wizard**

3. What file extension is typically used for Entity Data Models?

- A) .cs
- B) .xml
- C) .edmx
- D) .config
- **Answer: C) .edmx**

4. How can developers customize the generated classes in an Entity Data Model?

- A) By modifying the .edmx file directly
- B) By using partial classes
- C) By editing the database schema

- D) By changing the connection string
- **Answer: B) By using partial classes**

5. Which method is commonly used to save changes to the database in the Entity Framework?
    - A) SaveChanges()
    - B) Commit()
    - C) Update()
    - D) ApplyChanges()
    - **Answer: A) SaveChanges()**
6. What language is often used for querying data in the Entity Framework?
    - A) SQL
    - B) LINQ (Language-Integrated Query)
    - C) JSON
    - D) XML
    - **Answer: B) LINQ (Language-Integrated Query)**
7. What method is used to retrieve all records from a table using LINQ?
    - A) SelectAll()
    - B) FetchAll()
    - C) All()
    - D) ToList()
    - **Answer: D) ToList()**
8. What is an anonymous type in C#?
    - A) A class with no name
    - B) A type that is defined without specifying a class name
    - C) A type that stores dynamic data
    - D) A type that is defined at runtime
    - **Answer: B) A type that is defined without specifying a class name**
9. How do you create an anonymous type in C#?
    - A) var obj = new { Name = "John", Age = 30 };
    - B) var obj = new AnonymousType { Name = "John", Age = 30 };
    - C) var obj = AnonymousType();
    - D) var obj = new object { Name = "John", Age = 30 };
    - **Answer: A) var obj = new { Name = "John", Age = 30 };**
10. What method can be used to force the execution of a LINQ query?
    - A) Execute()
    - B) Fetch()
    - C) ToList()
    - D) Run()
    - **Answer: C) ToList()**
11. What does XAML stand for?
    - A) Extensible Application Markup Language

- B) Extensive Application Markup Language
- C) Extended Application Markup Language
- D) Extensible Array Markup Language
- **Answer: A) Extensible Application Markup Language**

12. Which of the following is a common control in XAML?

- A) Button
- B) Timer
- C) Socket
- D) Stream
- **Answer: A) Button**

13. How do you set a property of a control in XAML?

- A) By using a constructor
- B) By using attribute syntax
- C) By using a method
- D) By using a lambda expression
- **Answer: B) By using attribute syntax**

14. Which attribute is used to handle a button click event in XAML?

- A) Click
- B) OnClick
- C) HandleClick
- D) ButtonClick
- **Answer: A) Click**

15. Which layout control in XAML arranges its children into a single line either horizontally or vertically?

- A) StackPanel
- B) Grid
- C) Canvas
- D) DockPanel
- **Answer: A) StackPanel**

16. What is the main context class used to interact with the database in the Entity Framework?

- A) DataContext
- B) DbContext
- C) SqlContext
- D) QueryContext
- **Answer: B) DbContext**

17. Which component of the Entity Data Model represents a collection of entities?

- A) EntitySet
- B) EntityType
- C) EntityCollection

- D) EntityGroup
- **Answer: A) EntitySet**

18. How do you perform a simple LINQ query to get all records where the name is "John"?

- A) context.People.Where(p => **p.Name** == "John").ToList()
- B) context.People.Select(p => **p.Name** == "John").ToList()
- C) context.People.Find(p => **p.Name** == "John").ToList()
- D) context.People.Fetch(p => **p.Name** == "John").ToList()
- **Answer: A) context.People.Where(p => p.Name == "John").ToList()**

19. Which XAML layout control provides a flexible grid area that consists of columns and rows?

- A) Grid
- B) StackPanel
- C) Canvas
- D) Border
- **Answer: A) Grid**

20. How do you attach an event handler to a control in XAML?

- A) By setting the event property to the handler name in the XAML file
- B) By calling the event handler directly
- C) By overriding the event method
- D) By using a delegate
- **Answer: A) By setting the event property to the handler name in the XAML file**

1. What is data binding in the context of application development?

- A) The process of formatting data
- B) The process of connecting UI elements to data sources
- C) The process of compiling code
- D) The process of encrypting data
- **Answer: B) The process of connecting UI elements to data sources**

2. Which XAML attribute is used to bind a control to a data source?

- A) DataSource
- B) Binding
- C) DataContext
- D) Source
- **Answer: C) DataContext**

3. How do you bind a TextBlock control to a property named "Name" in XAML?

- A) `<TextBlock Text="{Binding Path=Name}" />`
- B) `<TextBlock Text="{DataContext=Name}" />`
- C) `<TextBlock Text="{Source=Name}" />`
- D) `<TextBlock Text="{Name=Binding}" />`
- **Answer: A)** `<TextBlock Text="{Binding Path=Name}" />`

4. Which method is used to set the data context of a control in code-behind?
   - A) SetDataContext()
   - B) SetBinding()
   - C) DataContext()
   - D) BindingContext()
   - **Answer: A) SetDataContext()**
5. How do you bind a ListBox control to a collection in XAML?
   - A) `<ListBox ItemsSource="{Binding}" />`
   - B) `<ListBox DataSource="{Binding}" />`
   - C) `<ListBox Collection="{Binding}" />`
   - D) `<ListBox Items="{Binding}" />`
   - **Answer: A)** `<ListBox ItemsSource="{Binding}" />`
6. Which class in .NET is commonly used for creating tasks?
   - A) Task
   - B) Thread
   - C) Process
   - D) ThreadPool
   - **Answer: A) Task**
7. How do you start a new task in .NET?
   - A) `Task.Run(() => { /* code */ });`
   - B) `new Thread(() => { /* code */ }).Start();`
   - C) `Process.Start(() => { /* code */ });`
   - D) `ThreadPool.QueueUserWorkItem(() => { /* code */ });`
   - **Answer: A) `Task.Run(() => { /* code */ });`
8. How can you wait for a task to complete before proceeding?
   - A) `task.Wait();`
   - B) `task.Join();`
   - C) `task.Hold();`
   - D) `task.Sleep();`
   - **Answer: A) `task.Wait();`
9. Which type is used to create a task that returns a value?
   - A) Task<T>
   - B) ValueTask
   - C) Task
   - D) FutureTask
   - **Answer: A) Task<T>**
10. Which token is used to support task cancellation?
   - A) CancellationToken
   - B) StopToken
   - C) EndToken
   - D) AbortToken

- **Answer: A) CancellationToken**

11. How do you signal a task to cancel using a CancellationTokenSource?
    - A) `tokenSource.Cancel();`
    - B) `tokenSource.Stop();`
    - C) `tokenSource.Abort();`
    - D) `tokenSource.End();`
    - **Answer: A) `tokenSource.Cancel();`

12. Which method is used to run multiple tasks in parallel?
    - A) `Task.Run()`
    - B) `Task.WhenAll()`
    - C) `Task.Parallel()`
    - D) `Task.MultiRun()`
    - **Answer: B) `Task.WhenAll()`

13. How do you chain tasks to run one after the other?
    - A) `task.ContinueWith(nextTask);`
    - B) `task.Then(nextTask);`
    - C) `task.Follow(nextTask);`
    - D) `task.Chain(nextTask);`
    - **Answer: A) `task.ContinueWith(nextTask);`

14. What is the default binding mode for a TextBox's Text property?
    - A) OneWay
    - B) TwoWay
    - C) OneTime
    - D) Default
    - **Answer: B) TwoWay**

15. How do you bind a ComboBox control to a collection in XAML?
    - A) `<ComboBox ItemsSource="{Binding}" />`
    - B) `<ComboBox DataSource="{Binding}" />`
    - C) `<ComboBox Collection="{Binding}" />`
    - D) `<ComboBox Items="{Binding}" />`
    - **Answer: A) `<ComboBox ItemsSource="{Binding}" />`

16. How do you set the data context of a control in code-behind?
    - A) `control.DataContext = data;`
    - B) `control.SetBinding(data);`
    - C) `control.SetDataContext(data);`
    - D) `control.DataSource = data;`
    - **Answer: A) `control.DataContext = data;`

17. How do you create a task that runs a method called "DoWork"?
    - A) `Task.Run(() => DoWork());`
    - B) `Task.Start(() => DoWork());`
    - C) `Task.Begin(() => DoWork());`

- D) `Task.Execute(() => DoWork());`
- **Answer: A) `Task.Run(() => DoWork());`

18. How do you access the result of a task that returns a value?

- A) `task.Result`
- B) `task.GetResult()`
- C) `task.FetchResult()`
- D) `task.ReturnValue`
- **Answer: A) `task.Result`

19. How do you pass a CancellationToken to a task?

- A) `Task.Run(action, token);`
- B) `Task.Create(action, token);`
- C) `Task.New(action, token);`
- D) `Task.Start(action, token);`
- **Answer: A) `Task.Run(action, token);`

20. Which method is used to wait for any one of multiple tasks to complete?

- A) `Task.WhenAny()`
- B) `Task.WhenAll()`
- C) `Task.WaitAny()`
- D) `Task.WaitAll()`
- **Answer: A) `Task.WhenAny()`